NUS ACE YOUTH PROGRAMME

AI AND MACHINE LEARNING

.Programme Content

	Outline
Day 1	- Introduction to AI o A brief review on AI history o AI Applications: State of the art o AI, Machine Learning, and Deep Learning o Restrictions and constraints o Future of AI - Introduction to Orange/AI Studio/Google Colab o Why Orange/AI Studio/Google Colab? o Setting up your system o Creating Your First Workflow/Program - Introduction to Python (Preprocessing of Data) o Checking and handling missing values o Handling categorical data (Label encoding, One-hot encoding) o Standardize/Normalize continuous data o PCA transformation o Data splitting Continuous Assessment MCQ Quiz 1
Day 2	- Machine Learning Methods Using Orange/AI Studio/Google Colab o Importing the data files o Understanding our Data o Missing Values and Imputation o Training your First Model o Supervised Learning Algorithms ▪ Logistic Regression ▪ Random Forest ▪ Support Vector Machine ▪ K-Nearest Neighbor ▪ Neural Network ▪ Comparison of Different Models o Clustering Algorithms ▪ Hierarchical Clustering ▪ K-Means o Visualization ▪ Data Table ▪ Scatter Plot ▪ Mosaic Display ▪ Sieve Diagram ▪ Rank ▪ Rad Viz Continuous Assessment MCQ Quiz 2
Day 3	Machine Learning Methods Using Python/Google Colab

	- Clustering and its applications o Clustering ▪ Definition of Clustering Task ▪ Popular Clustering Algorithms: K-means, Hierarchical, DBSCAN o Applications ▪ Customer segmentation, Image segmentation, Anomaly detection - Decision Tree and its applications o Decision Tree ▪ Definition of Decision Tree task ▪ Explain decision tree method o Applications ▪ Credit risk assessment, Disease diagnosis, Predictive maintenance Continuous Assessment MCQ Quiz 3
Day 4	- Machine Learning Case Studies and Application o Using Orange/AI Studio/Google Colab ▪ Healthcare: Predicting Disease Outcomes (CLASSIFICATION) ▪ Finance: Fraud Detection (OUTLIER DETECTION) ▪ Marketing: Customer Segmentation (CLUSTERING) - Machine Learning Case Studies and Application o Using Orange/AI Studio/Google Colab ▪ Retail: Personalized Recommendations (RECOMMENDER SYSTEMS) - LLM (Large Language Model) Applications - End Term Assessment MCQ Quiz 4
Day 5	Final Group Presentation and Assessment